HARARE INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION SCIENCES AND TECHNOLOGY

B.TECH (HONS) DEGREE IN INFORMATION SECURITY AND ASSURANCE

PART 2 SEMESTER 1

ISS2103: DATA COMMUNICATION AND NETWORKS

TIME: 3 HOURS **DATE: NOV/ DEC 2024**

INSTRUCTIONS TO CANDIDATES:

- *This paper comprises five Questions.*
- *Answer any four questions.*
- *Each question carries 25 marks*
- *Mark value to each sub parts of question are shown in right hand side.*
- *Illustrate your answer, where appropriate with large clearly labelled diagrams.*
- *Start a new question on a fresh page*

ADDITIONAL MATERIALS

- *None*

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Question 1

a) Draw the typical structure of an IP header and state the purpose of the following fields [2]
   i. Time to live. [2]
   ii. Protocol. [3]
   iii. Source and Destination IP addresses. [6]
b) Explain the concept of network segmentation.
c) The data link layer in the IEEE standard is divided into two (2) sub layers. Explain the layers in detail. [6]
d) You have sub-netted your class C network 192.168.1.0 with a subnet mask of 255.255.255.240. Calculate and list the following:
   i. number of networks. [1]
   ii. number of hosts per network. [1]
   iii. the full range of the first three networks [2]
   iv. the usable address ranges from those first three networks. [2]

Question 2

a) VPN can be essential in network security, briefly explain how VPN works. [3]
b) Discuss the concept of IP addressing and subnetting. [5]
c) Identify and explain any three (3) types of network topologies. [9]
d) In data communications and networking, switching is the practice of directing a signal or data element toward a particular hardware destination.
   i. Explain the main difference between a circuit-switched network and a packet switched network. [4]
   ii. Give two (2) examples of packet-switched protocols and two (2) examples of circuit switched protocols. [4]

Question 3

a) Express the following IPv6 addresses in shortened form.

   2001:0DB8:0000:1470: 0000:0000:0000:0300 [1]

b) Describe the differences between LAN, WAN, and MAN. [9]
c) With the aid of a clearly labelled diagram, define the OSI model and explain its layers. [15]

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Question 4

a) With the aid of a supporting diagram, Explain the concept of Carrier Sense Multiple Access/Collision Detection mechanism used by Ethernet (CSMA/CD). [5]

b) Elaborate on software-defined networking (SDN). [5]

c) In the Aloha system, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel. Explain using a diagram the functionality of:

   i. Pure Aloha. [4]

   ii. Slotted Aloha. [4]

d) A large enterprise network is experiencing slow data transfer rates, high latency and frequent packet losses. The network administrator is tasked with improving the network's performance to ensure efficient and reliable communication among employees, customers and partners.

Describe network performance optimization and how would you achieve it in this scenario to improve the network's efficiency, speed and reliability? [7]

Question 5

a) In contrast to TCP, UDP is described as a connectionless protocol. Briefly explain how data is transmitted between two computers using the UDP protocol. [4]

b) A common method for providing broadband Internet access over existing telephone lines is Asymmetric Digital Subscriber Line (ADSL).

   i. Explain why is it called asymmetric? [2]

   ii. Describe how ADSL is able to transmit both data and telephone calls over the same twisted pair cable connecting a house to a local exchange, without the two signals interfering with each other. [5]

c) Explain in detail the working of the RIP v2 routing algorithm. [6]

d) The Global unicast IPv6 addresses can be assigned dynamically in two different ways: stateless address auto configuration (SLAAC) and dynamic host configuration protocol v6 (DHCPv6). Describe the differences between the two methods. [8]

.....END OF EXAMINATION PAPER.....

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B. TECH (HONS) DEGREE IN INFORMATION SECURITY AND ASSURANCE

PART 2 SEMESTER 1

ISS 2101: SECURE CODING

TIME: 3 HOURS **DATE: NOV/DEC 2024**

TOTAL MARKS: 100

INSTRUCTIONS TO CANDIDATES:

- This question paper contains a total of **FIVE** questions.
- Each question carries **25** marks
- Answer **any FOUR** questions
- Start a new question on a fresh page

ADDITIONAL MATERIAL:

- None

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QUESTION 1

a) Briefly explain the following terms in cyber security:

i Byte Code. [1]

ii CSRF. [1]

b) Write a Java Class to defend against an SQL Injection attack. [8]

c) You have recently joined the IT Department of a local startup as a DevOps Engineer. You are part of a team of developers working on a Taxi Booking Application. Explain the security practises that you would consider during the design, implementation and deployment of the application. [15]

QUESTION 2

a) Explain the difference between session attacks and cookie attacks. [4]

b) Discuss the key ways in which data types affect the security of an application program. [5]

c) Discuss the Saltzer and Schroeder (1975)'s classic security design principles. [7]

d) Java has a robust security architecture designed to protect against various threats. Describe these security characteristics. [9]

QUESTION 3

a) Discuss the advantages of OOP in mitigating security vulnerabilities. [10]

b) Explain the following terms in secure coding:

i Encapsulation. [2]

ii Inheritance. [2]

iii Polymorphism. [2]

iv Interfaces. [2]

c) Discuss the common Java exception errors. [7]

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QUESTION 4

a) Elucidate the critical security risks you should consider for a web application. [20]

b) Explain the use of access modifiers (public, private, protected) in Java and their role in implementing secure coding. [5]

QUESTION 5

a) Discuss the significance of secure exception handling and error reporting in the CIA triad. [10]

b) Explain five (5) security attacks on the Confidentiality of a web applications. [10]

c) Write a Java code that is vulnerable to Broken Access Control according to the OWASP Top 10. [5]

**********END OF PAPER**********

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SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

DEPARTMENT OF INFORMATION SECURITY AND ASSURANCE

B. TECH (HONS) DEGREE IN INFORMATION SECURITY AND ASSURANCE

PART 2 SEMESTER 1

ISS 2104: CRYPTOGRAPHY AND SECURITY

TIME: 3 HOURS **DATE: NOVEMBER/DECEMER 2024**

TOTAL MARKS: 100

INSTRUCTIONS TO CANDIDATES:

- This question paper contains a total of **FIVE** questions.
- Each question carries **25** marks
- Answer **any FOUR** questions
- Start a new question on a fresh page

ADDITIONAL MATERIAL:

- Scientific Calculator

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QUESTION 1

a) Differentiate between the following terms clearly:

i) Cryptography and steganography. [2]

ii) Block cipher and stream cipher. [2]

iii) Cryptanalysis and cryptology. [2]

b) Distinguish between symmetric and asymmetric cryptosystems and give advantages and disadvantages of each category. [8]

c) Explain the different techniques used to classify crypto systems and highlight the ciphers under each technique. [11]

QUESTION 2

a) Discuss three **(3)** block cipher modes of operation using diagrams. [12]

b) Explain at least five **(5)** techniques that an organisation can use to protect its information assets from various attacks. [10]

c) Differentiate between http and https. [3]

QUESTION 3

a) Describe the RSA (Rivest Shamir and Adelman) public key cryptosystem. Your answer should include:

i. The generation of public and private keys. [5]

ii. The encryption algorithm. [3]

iii. Decryption algorithm. [2]

iv. The basis for the security of RSA. [2]

b) Bob has a public **key (n =65, e=5),** show that Bob's private key ***d*** is **29** [5]

c) Alice wants to send message **m =11** to Bob. She encrypts the message using Bob's public key. What is the value of the cipher text that Alice sends to Bob? [3]

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d) David has also sent an encrypted message to Bob. The cipher text value that Bob received from David is **19**. Showing all your working, use Bob's key to decrypt this cipher text and recover the value of David's message. **[6]**

QUESTION 4

a) Write down the general equations for encryption and decryption using Vigenere cipher. **[2]**

b) Using the Vigenere cipher, perform encryption on plain text "***we are discovered hide yourself***" using a key "***cryptography***". Show all your working. **[8]**

c) Encrypt the following text "***Welcome to cryptography and security***" using column transposition cipher with the key: "***security***". **[6]**

d) Perform decryption on the cipher text in (**c**) to get the plain text. Show all your workings. **[6]**

e) Outline the differences between computationally secure and unconditionally secure principles of cryptographic algorithms. **[3]**

QUESTION 5

a) List any five **(5)** components of a digital certificate. **[5]**

b) Explain the components of a Public Key Infrastructure. **[10]**

c) Describe the properties of a hashing function. **[10]**

***********************END OF PAPER************************

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Question 1

a) Explain information risks in project management and ways that they can be mitigated. [9]

b) Describe how security can be incorporated in each of the following phases of Software Development Life Cycle (SDLC).

i. Initiation. [4]

ii. Development/Acquisition. [4]

iii. Implementation. [4]

iv. Operation and Maintenance. [4]

Question 2

a) "One of the true tests of a leadership is the ability to recognize the problem before it becomes an emergency" (Arnold Gaslow). Critically analyze the statement above outlining the roles of a project leader. [10]

b) "Practice the 20/80 rule the first 20 % of the time you spend planning a project will be worthy 80% of the time you spend on the entire project". Discuss the use Pareto's principle (80/20) to help in secure software management. [10]

c) Describe the strategic considerations to be used before authorizing a project. [5]

Question 3

a. Project risk management is the process of identifying, analyzing and responding to project risk. Explain the types of project risks. [9]

b. Explain ways to communicate risks effectively to stakeholders. [4]

c. Elaborate the following terms in respect to secure project management. [4]

i. Qualitative risks analysis. [4]

ii. Quantitative risk analysis. [4]

iii. Risk monitoring and control. [6]

Question 4

a) Discuss the basic principles of project scheduling

b) Table 1 is a precedence table for installing a new accounting system for an organization.

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Table 1: Precedence Table

Activity	Precedents	Duration (weeks)
A		6
B		4
C	A	3
D	B	4
E	B	3
F		11
G	E,F	4
H	C,D	2

i. Draw an activity on arrow diagram to depict the information on the table. [8]

ii. Calculate the early and late times for all the activities [8]

iii. Identify the critical path [2]

iv. Calculate the free slack on all the nodes which are not on the critical path [8]

Question 5

a) Discuss the steps involved in step wise planning. [12]

b) The Delphi technique is a human based estimation technique. Explain the eight (8) basic steps of the Delphi technique. [8]

c) Define Work Breakdown Structure (WBD) and how does it affect the work estimates of tasks/activities. [5]

END OF EXAMINATION

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